



DET NORSKE VERITAS

DESIGN VERIFICATION REPORT

Client : Special and Safety Work

DNV Project No : PP057041

Equipment Description : Steel Security Window (2000mmx 1200mm)

Client Project : MOI Project

Design Code : AISC (9th Edition)

Referred documents (as provided by SSW):

- (1) Drawing No. DNV-01 Rev. 2 titled Security Window with Steel Bars (Elevation and Section) – attached as Annexure-1 with DVR.
- (2) Intertek Test Report of Air Infiltration and Water Penetration as Annexure -2 of the DVR.
- (3) Silicone Sealant to be used is Tremco Spectrem-2, Class-50, Product Data Sheet – attached as Annexure-3 of the DVR.
- (4) Glazing tape is 1/8" V2104 Thermalbond- attached as Annexure-3 of the DVR.
- (5) Elbex Neoprene setting block: Die No. 2787- attached in Annexure-3 of teh DVR.
- (6) Glazing report- 1/2" Makralon solid polycarbonate sheet – product data sheet- attached in Annexure-3 of DVR.
- (7) Inspection report for 1" tool resistant steel bar as Annexure-4 of this report.
- (8) Steel Tube Material certificate as Annexure-4 of this report.

This Is To Certify That:

Design of the Steel security window (2000mm x 1200mm) – drawing No. DNV-01 Rev. 0 has been reviewed and the design complies with the Design Code with regards to local in-place strength against uniformly distributed static equivalent pressure load of 98 kg/m².

Validity Period:

No validity period is stated on the design verification report as it only confirms the conformity with specified regulations / standards at the time of issue. Change in regulations or change in design may have influence on the validity of this report.

Scope of Verification:

Verification of the design of Window is limited to review of structural analysis submitted for compliance with AISC (9th Edition). No other codes or standards are referred to. Client to satisfy themselves regarding the requirements of other relevant codes / standards / national or local regulations if applicable.

If any person suffers loss or damage which is proved to have been caused by any negligent act or omission of Det Norske Veritas, then Det Norske Veritas shall pay compensation to such person for his proved direct loss or damage. However, the compensation shall not exceed an amount equal to ten times the fee charged for the service in question, provided that the maximum compensation shall never exceed USD 300,000 USD. In this provision "Det Norske Veritas" shall mean the Foundation Det Norske Veritas as well as all its subsidiaries, directors, officers, employees, agents and any other acting on behalf of Det Norske Veritas.

This review does not cover the design and integrity of concrete structure around the window. The suitability of the window components against fire resistance, the strength and quality of the window glass is not part of the present scope. The validity of the reports (glazing, sealant and tests, including air and water penetration) and certificates are not being verified by DNV. The in-service deterioration of the window components due to various causes like vibration, thermal expansion, contraction, etc. is not examined by DNV as part of this report.

Below mentioned parameters are considered in review –

Maximum Pressure load	98 kg/m ²
Overall frame size	2000mm width and 1200mm height

The below test parameters are used for reference only and not subject to approval:

Test Number:	1 (Pre-application)	2 (Post-application)
Test Pressure (PSF):	1.57	1.57
Sample Leakage (SCFM):	3.41	0.16
Leakage per Square Foot		
Sample Area (SCFM/ft ²):	0.36	0.02

List of Documents:

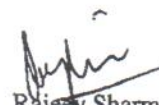
Document Title	Document No.	Rev.	Status
Security Window with Steel Bars	DNV-01	2	Approved with comment.

Comments:

1. The design verification is limited to review of the structural part (frame only) and within the scope of verification listed above.
2. SSW to ensure that the fabrication, installation and adjustment of various components of the window will be as per manufacturer's recommendations and should be able to meet the performance requirements stipulated by the client.
3. Reference to SSW drawing, it is understood that Impact loading is not applicable hence the same is not considered in the design verification.
4. Method of installation of window components are not included in the review.

Signed for Det Norske Veritas, Abu Dhabi


Jens Philip
Engineer


Rajeev Sharma
Service Lead- Design Appraisal

Date: 17th December 2012
Place: Abu Dhabi, UAE